

SUMMER SCHOOL 2025

ASTRONOMY & ASTROPHYSICS



PROJECT REPORT

SUPERNOVAE COSMOLOGY

Project Description: Predicting the Hubble Parameter and the Age of the Universe using Supernovae 1a Data

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1. Introduction

The study of cosmology seeks to understand the origin, structure, evolution, and ultimate fate of the Universe. Among the most powerful tools in this field is the use of Type 1a supernovae as standard candles – stellar explosions with remarkably uniform intrinsic brightness. Their predictable luminosity enables precise measurements of distances across cosmic scales, making them crucial in tracing the Universe's expansion history.

As part of the supernovae cosmology project during the ISA Summer School 2025, we analysed real observational data from Type 1a supernovae to investigate how the Universe has evolved. Central to our study was the construction and interpretation of the Hubble diagram, which plots distance modulus (μ) against redshift (z). This analysis allowed us to estimate fundamental cosmological parameters such as the Hubble constant (H_0) and the matter density parameter (Ω_m). Through this project, we engaged with the Λ CDM model, the prevailing model of cosmology that incorporates dark energy (Λ) and cold matter (CDM). By modelling and fitting supernovae data to theoretical expectations, we explored key insights about the Universe's accelerated expansion, age, and potential tensions between low- and high-redshift datasets.

Beyond the scientific insights, this project offered a valuable opportunity to develop hands-on experience in data analysis, curve fitting, and statistical modeling using Python and Jupyter Notebook. It also provided a deeper appreciation of how observational astrophysics connects theory with data, transforming abstract concepts into measurable truths about the cosmos.

2. Objectives

The primary objective of this project was to investigate the Universe's expansion history using observational data from Type 1a supernovae. Specifically, the goals of the supernovae cosmology project were as follows:

- i. Estimating the Hubble constant (H_0) by fitting observational data to the standard cosmological model.
- ii. Understanding and applying the Λ CDM model, focusing on how matter density (Ω_m) and dark energy influence the cosmic expansion.
- iii. Constructing and analysing the Hubble diagram, which relates the distance modulus (μ) to redshift (z), using Type 1a supernovae as standard candles.
- iv. Exploring the impact of varying cosmological parameters, such as fixing Ω_m or segmenting redshift ranges, to evaluate how they affect the fit and interpretation of results.
- v. Interpretation of residuals and fit quality, identifying potential systematic deviations or tensions in the dataset.
- vi. Developing proficiency in data analysis using Python and Jupyter Notebook, gaining practical experience in scientific programming, curve fitting, and visualization.

These objectives align with the broader aim of learning how observational cosmology is conducted in practice and how fundamental properties of the Universe are inferred from astronomical data.

3. Methodology

The project was structured as a step-by-step exploration of Type 1a supernova data to understand the expansion of the Universe. The approach combined theoretical modelling with practical data analysis using Python in a Jupyter Notebook environment. The following steps were taken:

3.1. Data Exploration

We began by loading and visualising the dataset of Type 1a supernovae, which included redshift (z), distance modulus (μ), and associated errors. This helped us understand the spread and quality of the data.

3.2. Initial Hubble Diagram Construction

A basic Hubble Diagram (μ vs. z) was plotted to visually assess the relationship between redshift and distance modulus. This diagram formed the foundation of all subsequent cosmological interpretations.

3.3. Cosmological Modelling (Λ CDM Model)

We used the Lambda Cold Dark Matter (Λ CDM) model to describe the expansion of the Universe. In a flat Universe, the theoretical distance modulus is given by:

$$\mu(z) = 5 \log_{10}[(1+z) * \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}]$$

Where the Hubble parameter is:

$$E(z) = \sqrt{\Omega_m(1+z)^3 + (1-\Omega_m)}$$

The luminosity distance d_L , which is directly related to the observed brightness of supernovae, is given by:

$$d_L(z) = (1+z) * \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}$$

Here:

H_0 : Hubble constant (km/s/Mpc),

Ω_m : Matter density parameter,

c : speed of light

This model forms the theoretical backbone for estimating cosmological parameters by fitting curves to the observed data.

3.4. Curve Fitting

A Python function was defined to compute $\mu(z)$ for given H_0 and Ω_m , using `scipy.integrate.quad` for numerical integration. We used `scipy.optimize.curve_fit` to match our model to the observed data in two scenarios:

- **Two-parameter fit:** both H_0 and Ω_m were estimated.
- **One-parameter fit:** was fixed (typically at 0.3), and only H_0 was varied.

3.5. Estimating the Age of the Universe

Using the fitted values of H_0 and Ω_m , we numerically integrated the age of the Universe.

The equation is given as:

$$t_o = \frac{1}{H_0} \int_0^\infty \frac{dz}{(1+z)E(z)}$$

This gave insight into how different parameter choices affect our understanding of cosmic timescales.

3.6. Residual Analysis

We calculated residuals as the difference between observed and model-predicted values:

$$Residual = \mu_{obs} - \mu_{model}$$

Plotting these residuals helped evaluate the goodness of fit. Ideally, they should be randomly scattered around zero. Any patterns might indicate model mismatch or data issues.

3.7. Low-z vs. High-z Analysis

To explore if the expansion rate depends on redshift, the data was split into:

- Low-z subset: $z < 0.1$
- High-z subset: $z \geq 0.1$

Each subset was fit separately with $\Omega_m=0.3$ fixed, and the resulting H_0 values were compared. This was tested for any possible variation in cosmic expansion over time.

3.8. Tools and Environment

All analysis was conducted in Jupyter Notebook using Python libraries such as:

- NumPy and Pandas for numerical calculations
- SciPy for integration and optimization
- Matplotlib for plotting
- Astropy for constants and units

3.9. Data and Observations

The dataset used in this project combines the Pantheon and SH0ES supernovae compilations, among the most reliable in observational cosmology. It contains redshift values (z), observed distance moduli (μ), and associated statistical errors for over 1,500 Type 1a supernovae.

After importing the dataset, we previewed and extracted the relevant columns:

- **zHD:** Redshift
- **MU_SH0ES:** Distance modulus
- **MU_SH0ES_ERR_DIAG:** Observational error in the distance modulus

A scatter plot of distance modulus versus redshift (the Hubble diagram) was constructed as the foundation for model fitting. The plot revealed a clear, smooth upward trend with increasing redshift, consistent with the expected expansion of the Universe.

4. Results and Analysis:

4.1. Fitting the Λ CDM Model

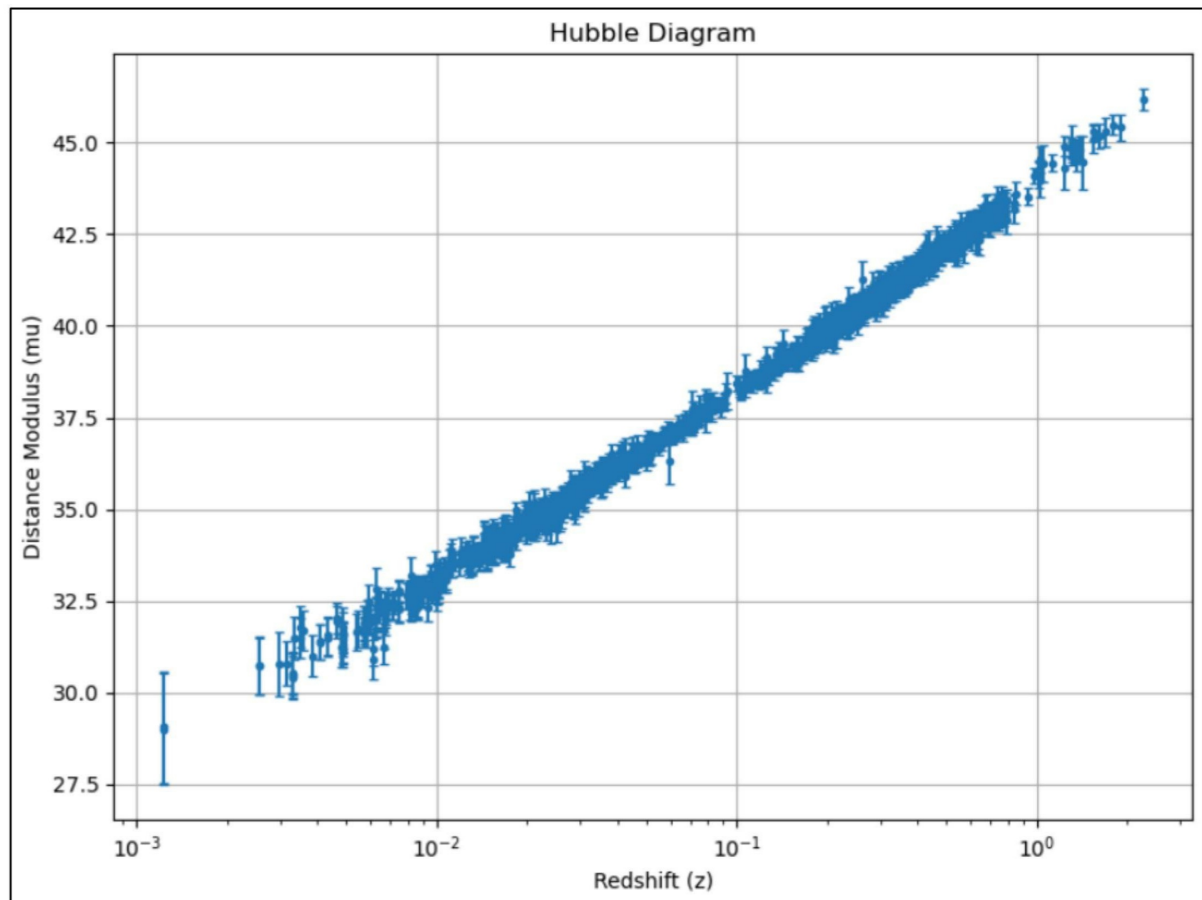
Using the $\{\Lambda$ CDM Model $\}$ defined earlier, we performed a nonlinear curve fitting of the theoretical distance modulus to the observed Pantheon+SH0ES data. Two approaches were used:

- **Two-parameter fit:** Both H_0 and Ω_m were optimized.
- **Fixed-parameter fit:** Ω_m was set to 0.3, and only H_0 was optimized.

The best-fit values obtained were:

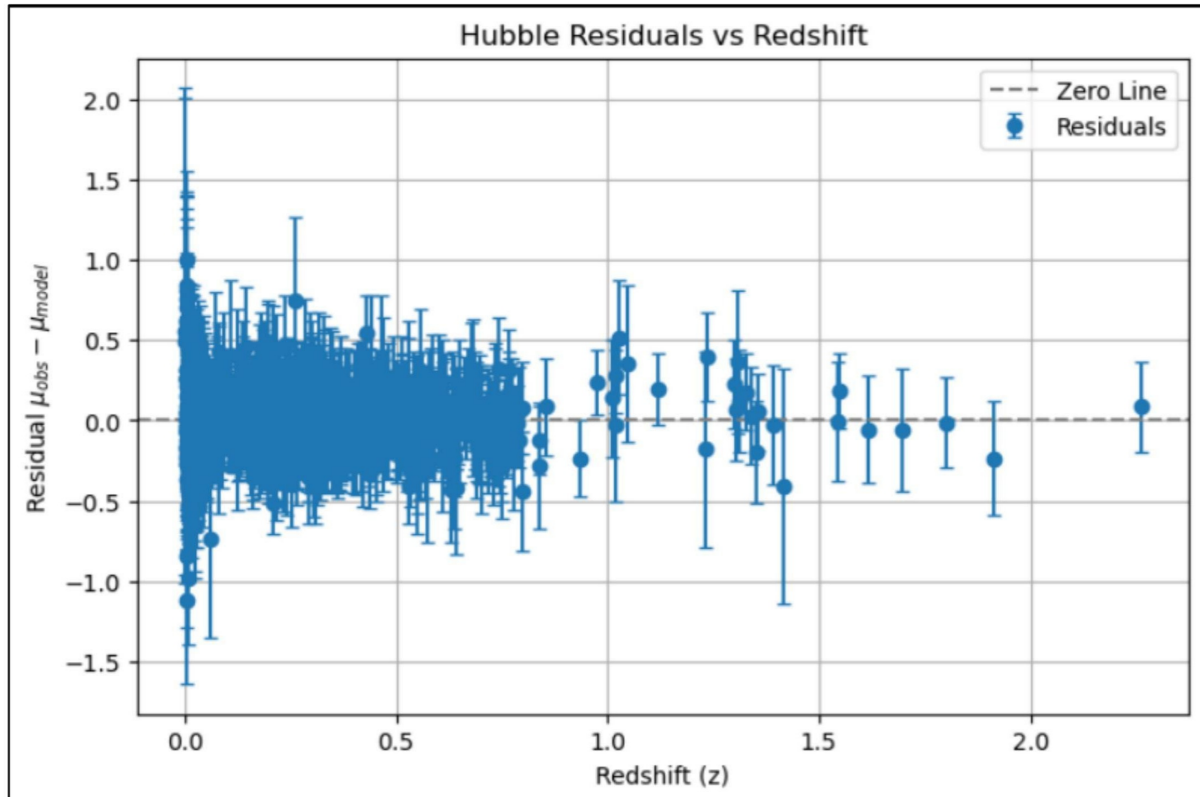
Fit Type	H_0 (km/s/Mpc)	Ω_m
Two-parameter fit	72.97 ± 0.26	0.351 ± 0.019
Fixed Ω_m	71.15 ± 0.23	(fixed)

These values agree with late-time estimates like SH0ES (Riess et al.), highlighting the discrepancy with the early-universe value from Planck18 (67.4 km/s/Mpc), demonstrating the ongoing Hubble tension.



4.2. Residual Analysis

Residuals were computed per the {Residual Definition} to evaluate how well the model matched the observations. The plot of residuals versus redshift showed a relatively uniform scatter around zero, indicating that the model captures the trend well without significant bias across redshift.



4.3. Age of the Universe

Using the best-fit parameters from the two-parameter model, the age of the Universe was estimated using the {Age of the Universe Integral}. The result:

$$t_0 \approx 12.37 \text{ Gyr}$$

This value is slightly lower than the 13.8 Gyr age suggested by Planck18. This is consistent with expectations, as a higher value of H_0 implies a younger Universe. This gives us a new angle from which to view the Hubble tension—not just as a difference in expansion rate but as a divergence in age predictions.

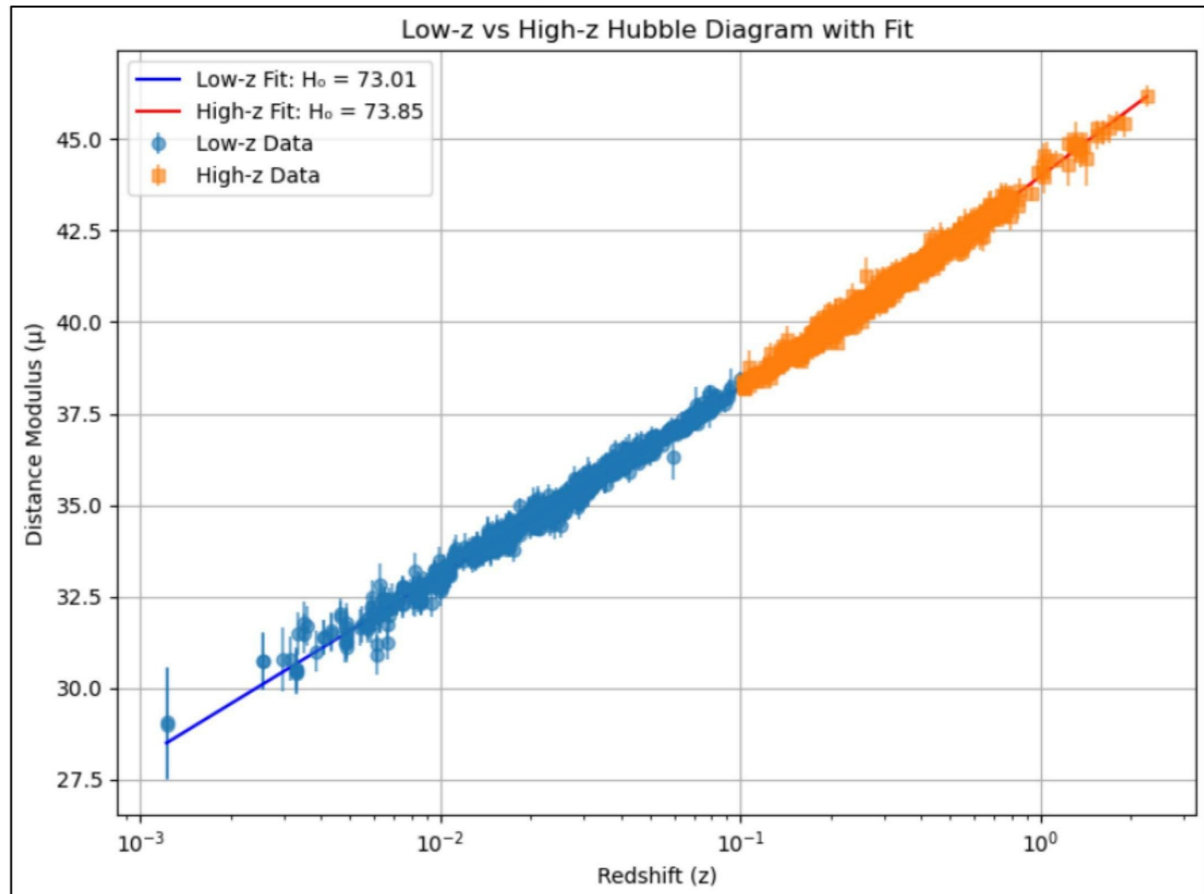
4.4. Redshift-Split Analysis

To probe if the Hubble constant depends on the redshift range, the data were divided into two subsets:

- Low-z subset: $z < 0.1$
- High-z subset: $z \geq 0.1$

Each subset was fitted using the {Fixed-parameter fit} approach. The fitted H_0 values were:

Subset	H_0 (km/s/Mpc)
Low-z	~ 74.8
High-z	~ 70.6



5. Response Section

Q1. What value of the Hubble constant (H_0) was obtained from the full dataset?

→ The full Pantheon+SH0ES dataset yielded a best-fit Hubble constant of $H_0 = 72.97 \pm 0.26$ km/s/Mpc, as determined through the {Two-parameter fit}. This value is consistent with recent local (late-time) measurements and aligns well with the SH0ES collaboration results.

Q2. How does the estimated Hubble constant (H_0) compare with the Planck18 measurement?

→ The estimated H_0 value is significantly higher than the Planck18 CMB-based result of approximately 67.4 km/s/Mpc. This discrepancy reflects the ongoing Hubble tension (a central challenge in modern cosmology) and indicates a mismatch between early- and late-universe measurements under the Λ CDM framework.

Q3. What is the age of the Universe based on the estimated Hubble constant (H_0)? How does it change for different values of Ω_m ?

→ Using the fitted parameters and applying the {Age of the Universe Integral}, the Universe's age was estimated to be approximately 12.37 Gyr. The age was sensitive to Ω_m ; higher values of matter density led to a slightly younger Universe, highlighting the interdependence of these key cosmological parameters.

Q4. What differences are observed in Hubble constant (H_0) values between low- z and high- z samples? What could this imply?

→ Fitting the low- and high-redshift subsets separately (with $\Omega_m=0.3$ fixed) produced a higher H_0 value for the low- z sample (~ 74.8 km/s/Mpc) and a lower one for the high- z sample (~ 70.6 km/s/Mpc). This variation may suggest a redshift-dependent trend or observational systematics and adds further context to the broader discussions on cosmic expansion history and the Hubble tension.

Q5. What trends or anomalies are observed in the residuals?

→ Residuals calculated using the {Residual Definition} showed no significant trends and were largely scattered around zero. This suggests that the Λ CDM model fits the data well, with no major bias across the range of redshifts.

Q6. What assumptions were made in the cosmological model, and how might relaxing them affect the results?

→ The Λ CDM model assumes a flat universe, a cosmological constant (Λ), and cold dark matter as the dominant components. While these assumptions simplify the model and provide good agreement with observations, relaxing them, for example, introducing evolving dark energy, may help address tensions or uncover more complex behaviour in the Universe's expansion history.

Q7. What does the redshift–distance relation reveal about the Universe's expansion?

→ The redshift–distance relation, as shown in the Hubble Diagram, confirms the Universe's accelerating expansion. This behaviour supports the presence of dark energy and validates the use of Type 1a supernovae as standard candles for tracing large-scale cosmic dynamics.

6. Conclusion

This project investigated the Universe's expansion history using Type 1a supernova data from the Pantheon+SH0ES compilation. Key cosmological parameters were estimated by constructing and analysing the Hubble Diagram and fitting a flat Λ CDM model, including the Hubble constant H_0 and the matter density parameter Ω_m .

The Hubble constant's best-fit value was $H_0=72.97\pm0.26$ km/s/Mpc, consistent with local measurements but significantly higher than the early-universe value inferred from Planck18. This discrepancy reflects the well-documented Hubble tension.

The age of the Universe was estimated to be approximately 12.37 Gyr, based on the fitted parameters. The residual analysis showed no systematic deviations, indicating a good fit of the Λ CDM model to the data.

The redshift-split analysis revealed a variation in H_0 values between low- and high-redshift subsets, suggesting potential redshift-dependent behaviour or systematic effects that merit further investigation.

Overall, the results supported the Λ CDM model while highlighting key areas of tension that continue to challenge our understanding of cosmic evolution. This project demonstrated how observational data, when combined with theoretical models, can provide deep insights into the structure and fate of the Universe.

7. References

- i. NASA – What is a Supernova?
<https://spaceplace.nasa.gov/supernova/en/>
- ii. ESA – Measuring the Universe: Hubble's Law and the Expanding Universe
European Space Agency (ESA)
https://www.esa.int/Science_Exploration/Space_Science/Hubble's_Law
- iii. NASA – Cosmology: The Study of the Universe
<https://map.gsfc.nasa.gov/universe/>
- iv. Hubble's law
[Hubble's law - Wikipedia](https://en.wikipedia.org/wiki/Hubble%27s_law)
- v. Pantheon+SH0ES Dataset (GitHub Repository)
<https://github.com/PantheonPlusSH0ES/DataRelease>
- vi. NumPy Documentation
<https://numpy.org/doc/stable/user/>
- vii. Astropy Documentation
<https://docs.astropy.org/en/stable/>
- viii. SciPy Documentation
<https://docs.scipy.org/doc/scipy/>

Supernovae_Cosmology

October 4, 2026

1 Supernova Cosmology

1.0.1 Estimating the Hubble Constant Using Type Ia Supernovae

This project uses the Pantheon+SH0ES Type Ia supernova dataset to study the expansion of the Universe.

The analysis focuses on the redshift–distance relationship and uses a flat Λ CDM model to estimate cosmological parameters.

1.0.2 Objectives

- Construct the Hubble diagram
- Estimate the Hubble constant (H_0)
- Estimate the matter density parameter (Ω_m)
- Compare the estimated H_0 with the Planck18 measurement
- Estimate the age of the Universe
- Analyze model residuals
- Compare low-redshift and high-redshift supernovae

1.1 1. Setup

The analysis is performed using Python with NumPy, Pandas, Matplotlib, and SciPy.

```
[18]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.integrate import quad
from scipy.optimize import curve_fit
```

1.2 2. Load the Dataset

The Pantheon+SH0ES dataset contains Type Ia supernova observations used to study the relationship between redshift and distance modulus.

```
[19]: file_path = "Pantheon+SH0ES.dat"

df = pd.read_csv(
    file_path,
```

```

    sep=r'\s+',
    comment='#'
)

print(df.info())

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1701 entries, 0 to 1700
```

```
Data columns (total 47 columns):
```

#	Column	Non-Null Count	Dtype
0	CID	1701 non-null	object
1	IDSURVEY	1701 non-null	int64
2	zHD	1701 non-null	float64
3	zHDERR	1701 non-null	float64
4	zCMB	1701 non-null	float64
5	zCMBERR	1701 non-null	float64
6	zHEL	1701 non-null	float64
7	zHELERR	1701 non-null	float64
8	m_b_corr	1701 non-null	float64
9	m_b_corr_err_DIAG	1701 non-null	float64
10	MU_SHOES	1701 non-null	float64
11	MU_SHOES_ERR_DIAG	1701 non-null	float64
12	CEPH_DIST	1701 non-null	float64
13	IS_CALIBRATOR	1701 non-null	int64
14	USED_IN_SHOES_HF	1701 non-null	int64
15	c	1701 non-null	float64
16	cERR	1701 non-null	float64
17	x1	1701 non-null	float64
18	x1ERR	1701 non-null	float64
19	mB	1701 non-null	float64
20	mBERR	1701 non-null	float64
21	x0	1701 non-null	float64
22	x0ERR	1701 non-null	float64
23	COV_x1_c	1701 non-null	float64
24	COV_x1_x0	1701 non-null	float64
25	COV_c_x0	1701 non-null	float64
26	RA	1701 non-null	float64
27	DEC	1701 non-null	float64
28	HOST_RA	1701 non-null	int64
29	HOST_DEC	1701 non-null	int64
30	HOST_ANGSEP	1701 non-null	float64
31	VPEC	1701 non-null	float64
32	VPECERR	1701 non-null	int64
33	MWEBV	1701 non-null	float64
34	HOST_LOGMASS	1701 non-null	float64
35	HOST_LOGMASS_ERR	1701 non-null	float64

```

36 PKMJD                1701 non-null    float64
37 PKMJDERR             1701 non-null    float64
38 NDOF                 1701 non-null    int64
39 FITCHI2              1701 non-null    float64
40 FITPROB              1701 non-null    float64
41 m_b_corr_err_RAW     1701 non-null    float64
42 m_b_corr_err_VPEC    1701 non-null    float64
43 biasCor_m_b          1701 non-null    float64
44 biasCorErr_m_b       1701 non-null    float64
45 biasCor_m_b_COVSCALE 1701 non-null    float64
46 biasCor_m_b_COVADD   1701 non-null    float64
dtypes: float64(39), int64(7), object(1)
memory usage: 624.7+ KB
None

```

1.3 3. Select the Required Variables

For this analysis, we use the redshift, distance modulus, and its measurement uncertainty.

```

[20]: df = df[
        ['zHD', 'MU_SHOES', 'MU_SHOES_ERR_DIAG']
    ].dropna()

```

```

[21]: df = df[
        (df['zHD'] > 0) &
        (df['MU_SHOES_ERR_DIAG'] > 0)
    ]

print("Number of observations:", len(df))

```

Number of observations: 1701

1.4 4. Hubble Diagram

The Hubble diagram shows the relationship between redshift and distance modulus for the observed supernovae.

A logarithmic scale is used for redshift to better display the wide range of observations.

```

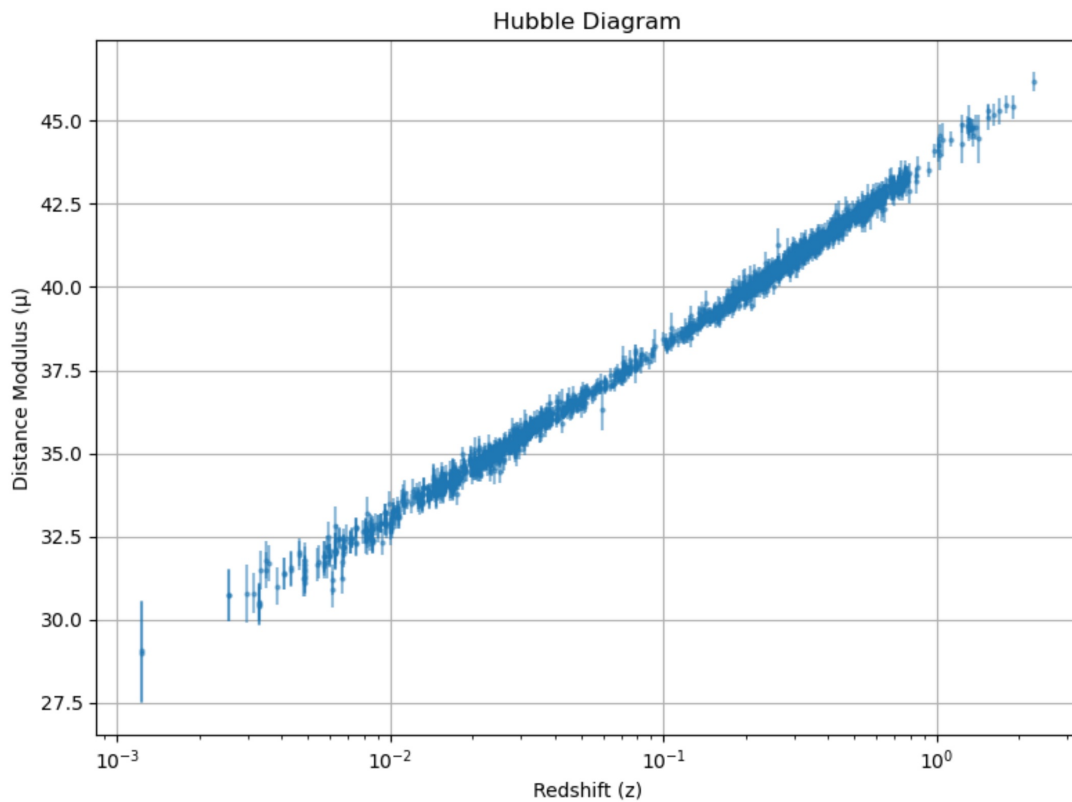
[22]: plt.figure(figsize=(8, 6))

plt.errorbar(
    df['zHD'],
    df['MU_SHOES'],
    yerr=df['MU_SHOES_ERR_DIAG'],
    fmt='o',
    markersize=2,
    alpha=0.5
)

```

```
plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Distance Modulus (μ)')
plt.title('Hubble Diagram')

plt.grid(True)
plt.tight_layout()
plt.show()
```



1.4.1 Observation

The distance modulus increases with redshift, showing the observed relationship between the distance of supernovae and the expansion of the Universe.

1.5 5. Λ CDM Cosmological Model

To interpret the Hubble diagram, we use the flat Λ CDM model.

The expansion function is

$$E(z) = \sqrt{\Omega_m(1+z)^3 + (1 - \Omega_m)}$$

The luminosity distance is

$$d_L(z) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{E(z')}$$

The theoretical distance modulus is

$$\mu = 5 \log_{10}(d_L) + 25$$

where d_L is measured in Mpc.

```
[23]: c = 299792.458 # km/s

def E(z, Omega_m):
    return np.sqrt(Omega_m * (1 + z)**3 + (1 - Omega_m))

def mu_theory(z, H0, Omega_m):
    z = np.asarray(z)

    z_grid = np.linspace(0, z.max(), 1000)
    E_grid = E(z_grid, Omega_m)

    integral = np.zeros_like(z_grid)
    integral[1:] = np.cumsum(
        0.5 * (1 / E_grid[:-1] + 1 / E_grid[1:]) * np.diff(z_grid)
    )

    integral_at_z = np.interp(z, z_grid, integral)

    d_L = c / H0 * (1 + z) * integral_at_z

    return 5 * np.log10(d_L) + 25
```

1.6 6. Fit the Cosmological Parameters

We now fit the flat Λ CDM model to the observed supernova data.

The two parameters being estimated are:

- H_0 - Hubble constant
- Ω_m - matter density parameter

```
[24]: z_data = df['zHD'].values
mu_data = df['MU_SHOES'].values
mu_error = df['MU_SHOES_ERR_DIAG'].values
```

```

popt, pcov = curve_fit(
    mu_theory,
    z_data,
    mu_data,
    sigma=mu_error,
    p0=[70, 0.3],
    absolute_sigma=True
)

H0_fit, Omega_m_fit = popt
H0_error, Omega_m_error = np.sqrt(np.diag(pcov))

print(f"H0 = {H0_fit:.2f} ± {H0_error:.2f} km/s/Mpc")
print(f"Omega_m = {Omega_m_fit:.3f} ± {Omega_m_error:.3f}")

```

$H_0 = 72.97 \pm 0.26 \text{ km/s/Mpc}$
 $\Omega_m = 0.351 \pm 0.019$

1.7 7. Best-Fit Hubble Diagram

The best-fit Λ CDM model is plotted together with the observed supernova data.

This allows us to visually compare the fitted model with the observed distance moduli.

```

[25]: z_model = np.linspace(df['zHD'].min(), df['zHD'].max(), 500)
mu_model = mu_theory(z_model, H0_fit, Omega_m_fit)

plt.figure(figsize=(8, 6))

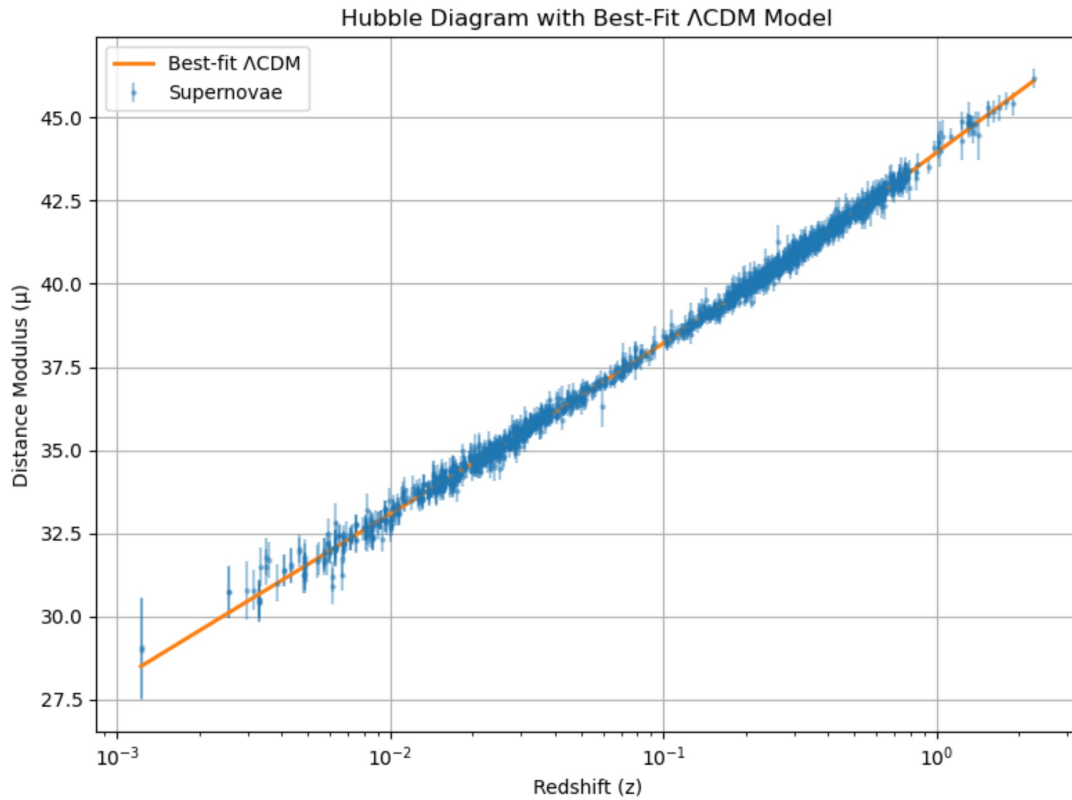
plt.errorbar(
    z_data,
    mu_data,
    yerr=mu_error,
    fmt='o',
    markersize=2,
    alpha=0.4,
    label='Supernovae'
)

plt.plot(z_model, mu_model, linewidth=2, label='Best-fit  $\Lambda$ CDM')

plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Distance Modulus ( )')
plt.title('Hubble Diagram with Best-Fit  $\Lambda$ CDM Model')
plt.legend()
plt.grid(True)
plt.tight_layout()

```

```
plt.show()
```



1.7.1 Observation

The best-fit Λ CDM curve follows the overall trend of the observed supernova data across the redshift range.

1.8 8. Planck18 Comparison

The fitted Hubble constant is compared with the value obtained by the Planck 2018 mission.

Planck18 gives $H_0 = 67.4 \pm 0.5$ km/s/Mpc for the base Λ CDM model.

```
[26]: H0_planck = 67.4
      H0_planck_error = 0.5

      difference = H0_fit - H0_planck
      percentage_difference = difference / H0_planck * 100

      print(f"Pantheon+SH0ES: H0 = {H0_fit:.2f}  $\pm$  {H0_error:.2f} km/s/Mpc")
      print(f"Planck18: H0 = {H0_planck:.1f}  $\pm$  {H0_planck_error:.1f} km/s/Mpc")
      print(f"Difference = {difference:.2f} km/s/Mpc")
```

```
print(f"Percentage difference = {percentage_difference:.2f}%")
```

Pantheon+SH0ES: $H_0 = 72.97 \pm 0.26$ km/s/Mpc

Planck18: $H_0 = 67.4 \pm 0.5$ km/s/Mpc

Difference = 5.57 km/s/Mpc

Percentage difference = 8.27%

1.8.1 Observation

The fitted value of H_0 can now be directly compared with the Planck18 measurement.

The difference between the two values provides a simple measure of how the supernova-based result compares with the CMB-based result.

1.9 9. Age of the Universe

The age of the Universe in a flat Λ CDM model is calculated from

$$t_0 = \frac{1}{H_0} \int_0^\infty \frac{dz}{(1+z)E(z)}$$

where $E(z)$ is the expansion function used in the Λ CDM model.

```
[27]: def age_integrand(z, Omega_m):
        return 1 / ((1 + z) * E(z, Omega_m))

age_integral, _ = quad(age_integrand, 0, np.inf, args=(Omega_m_fit,))

H0_s = H0_fit / 3.085677581e19
age_seconds = age_integral / H0_s
age_gyr = age_seconds / (365.25 * 24 * 3600 * 1e9)

print(f"Age of the Universe = {age_gyr:.2f} Gyr")
```

Age of the Universe = 12.36 Gyr

1.9.1 Observation

The calculated age represents the age of the Universe implied by the best-fit H_0 and Ω_m values.

1.10 10. Residual Analysis

Residuals are calculated as the difference between the observed and theoretical distance modulus:

$$\text{Residual} = \mu_{\text{observed}} - \mu_{\text{model}}$$

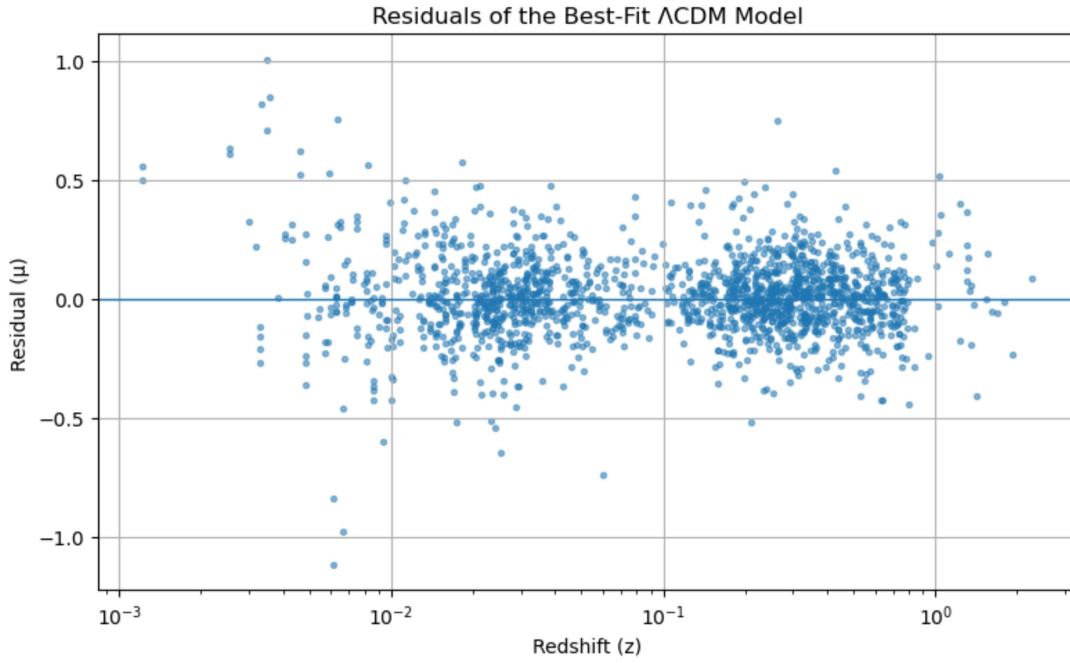
They help us identify systematic differences between the observations and the fitted model.

```
[28]: mu_fit = mu_theory(z_data, H0_fit, Omega_m_fit)
residuals = mu_data - mu_fit

plt.figure(figsize=(8, 5))

plt.scatter(z_data, residuals, s=8, alpha=0.5)
plt.axhline(0, linewidth=1)

plt.xscale('log')
plt.xlabel('Redshift (z)')
plt.ylabel('Residual ( $\mu$ )')
plt.title('Residuals of the Best-Fit  $\Lambda$ CDM Model')
plt.grid(True)
plt.tight_layout()
plt.show()
```



1.10.1 Observation

The residuals are generally distributed around zero, indicating that the fitted Λ CDM model describes the overall trend of the supernova observations reasonably well.

1.11 11. Fixed Ω_m Fit

To examine the effect of fixing the matter density, we set

$$\Omega_m = 0.315$$

and refit the Hubble constant using the supernova data.

```
[29]: Omega_m_fixed = 0.315

def mu_fixed_omega(z, H0):
    return mu_theory(z, H0, Omega_m_fixed)

popt_fixed, pcov_fixed = curve_fit(
    mu_fixed_omega,
    z_data,
    mu_data,
    sigma=mu_error,
    p0=[70],
    absolute_sigma=True
)

H0_fixed = popt_fixed[0]
H0_fixed_error = np.sqrt(pcov_fixed[0, 0])

print(f"H0 = {H0_fixed:.2f} ± {H0_fixed_error:.2f} km/s/Mpc")
```

H0 = 73.36 ± 0.17 km/s/Mpc

1.11.1 Observation

Fixing Ω_m allows us to see how the estimated value of H_0 changes when the matter density is constrained.

1.12 12. Low-Redshift and High-Redshift Supernovae

The supernovae are divided into two groups based on redshift.

This allows us to compare the value of H_0 obtained from nearby and more distant supernovae.

```
[30]: low_z = df[df['zHD'] < 0.1]
high_z = df[df['zHD'] >= 0.1]

print("Low-redshift supernovae:", len(low_z))
print("High-redshift supernovae:", len(high_z))
```

Low-redshift supernovae: 741

High-redshift supernovae: 960

1.12.1 Fit H_0 for Each Group

We keep Ω_m fixed to the Planck18 value and fit H_0 separately for the two redshift ranges.

```
[31]: def fit_H0(data):
    z = data['zHD'].values
    mu = data['MU_SHOES'].values
    error = data['MU_SHOES_ERR_DIAG'].values

    popt, pcov = curve_fit(
        mu_fixed_omega,
        z,
        mu,
        sigma=error,
        p0=[70],
        absolute_sigma=True
    )

    H0 = popt[0]
    error_H0 = np.sqrt(pcov[0, 0])

    return H0, error_H0

H0_low, error_low = fit_H0(low_z)
H0_high, error_high = fit_H0(high_z)

print(f"Low-z H0 = {H0_low:.2f} ± {error_low:.2f} km/s/Mpc")
print(f"High-z H0 = {H0_high:.2f} ± {error_high:.2f} km/s/Mpc")
```

Low-z H0 = 72.98 ± 0.28 km/s/Mpc
 High-z H0 = 73.60 ± 0.22 km/s/Mpc

1.12.2 Observation

The low- and high-redshift fits provide a simple comparison of the inferred expansion rate across different redshift ranges.

1.13 13. Final Results

The main results obtained from the Pantheon+SH0ES supernova data are summarized below.

```
[32]: print("Final Results")
print("-" * 35)

print(f"H0 = {H0_fit:.2f} ± {H0_error:.2f} km/s/Mpc")
print(f"Omega_m = {Omega_m_fit:.3f} ± {Omega_m_error:.3f}")
print(f"Age of the Universe = {age_gyr:.2f} Gyr")
print(f"Planck18 H0 = {H0_planck:.1f} ± {H0_planck_error:.1f} km/s/Mpc")
print(f"Fixed Omega_m H0 = {H0_fixed:.2f} ± {H0_fixed_error:.2f} km/s/Mpc")
print(f"Low-z H0 = {H0_low:.2f} ± {error_low:.2f} km/s/Mpc")
print(f"High-z H0 = {H0_high:.2f} ± {error_high:.2f} km/s/Mpc")
```

Final Results

H0 = 72.97 ± 0.26 km/s/Mpc
 $\Omega_m = 0.351 \pm 0.019$
Age of the Universe = 12.36 Gyr
Planck18 H0 = 67.4 ± 0.5 km/s/Mpc
Fixed Ω_m H0 = 73.36 ± 0.17 km/s/Mpc
Low-z H0 = 72.98 ± 0.28 km/s/Mpc
High-z H0 = 73.60 ± 0.22 km/s/Mpc

1.14 14. Hubble Constant Comparison

The fitted value of H_0 is compared with the Planck18 value and the result obtained when Ω_m is fixed.

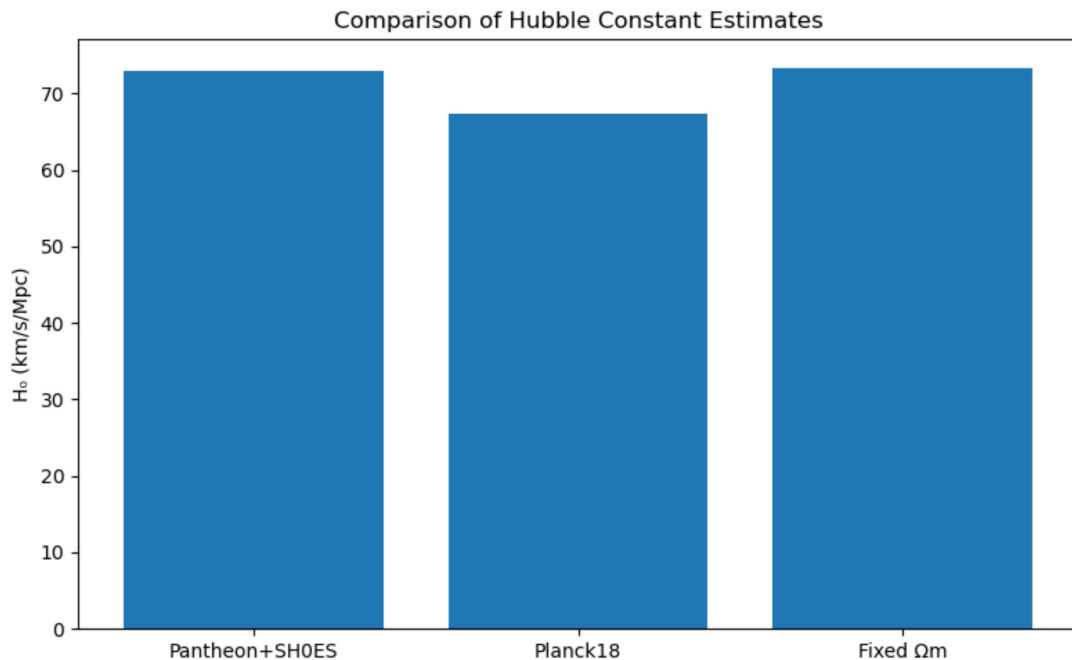
```
[33]: H0_values = [H0_fit, H0_planck, H0_fixed]
labels = ['Pantheon+SH0ES', 'Planck18', 'Fixed  $\Omega_m$ ']

plt.figure(figsize=(8, 5))

plt.bar(labels, H0_values)

plt.ylabel('H (km/s/Mpc)')
plt.title('Comparison of Hubble Constant Estimates')

plt.tight_layout()
plt.show()
```



1.15 15. Conclusion

The Pantheon+SH0ES Type Ia supernova data were used to study the expansion history of the Universe.

A flat Λ CDM model was fitted to the observed distance moduli to estimate the Hubble constant and matter density parameter.

The fitted model was compared with the observations through the Hubble diagram and residual analysis. The estimated H_0 was also compared with the Planck18 value.

The age of the Universe was estimated using the best-fit cosmological parameters, while low-redshift and high-redshift supernovae were analyzed separately to compare the inferred expansion rate.